Mitigating Hallucinations in Clinical Chatbots: A RAG Architecture for Family Planning

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***Abstract*— Patients seeking reproductive healthcare in the Philippines frequently face severe information barriers. Brief consultation times prevent clinic staff from explaining every available family planning option, including contraceptive eligibility and surgical screening rules. Consequently, unassisted patients often endure unnecessary clinic visits simply because they cannot independently decipher hundreds of pages of complex Department of Health guidelines. This research proposes a Retrieval-Augmented Generation (RAG) architecture to bridge this access gap. By mapping natural language queries directly to official medical manuals, a locally hosted language model synthesizes the correct passages into clear, accessible responses while strictly preserving data privacy. Establishing a quantitative evaluation framework ensures the system safely rejects out-of-scope prompts and avoids generating hallucinated medical advice. Ultimately, this architecture empowers patients to make informed, independent reproductive choices before arriving at a facility.**

Keywords—chatbot, clinical decision support, family planning, natural language processing, Philippines, retrieval-augmented generation

I. SELECTED UN SDG AND JUSTIFICATION

Achieving universal access to reproductive healthcare stands as a core objective under United Nations Sustainable Development Goal 3, specifically Target 3.7 [1]. The target mandates that nations ensure universal access to sexual and reproductive healthcare services, explicitly prioritizing family planning information and education. Translating that global mandate into local action across the Philippines remains difficult despite strong national legislation.

Patients, young couples, and mothers visiting rural community facilities frequently struggle to quickly verify their eligibility for specific methods, such as intrauterine devices (IUDs) or oral contraceptives, alongside broader family planning concerns like postpartum care, surgical screening prerequisites, and PhilHealth benefit entitlements. Such bottlenecks happen because the vital facts they need remain buried inside dense Department of Health clinical manuals and complex administrative circulars. Expecting the public to manually navigate extensive clinical guidelines and bureaucratic policies is highly impractical. A Retrieval-Augmented Generation architecture resolves this barrier by mapping natural language patient queries directly to the relevant clinical passages. Synthesizing those retrieved passages into coherent responses allows patients to instantly understand their eligibility, directly supporting the SDG mandate.

II. NARROW PROBLEM STATEMENT

National legislation, such as the Responsible Parenthood and Reproductive Health Act, guarantees family planning access and establishes strict clinical eligibility rules [2]. Primary care facilities integrate further compliance standards through the Universal Health Care Act [3]. For an unassisted patient, navigating these dense regulatory frameworks, such as understanding why a history of hypertension might disqualify a mother from receiving combined oral contraceptives under the WHO Medical Eligibility Criteria [4], is completely overwhelming.

While healthcare workers understand these clinical guidelines, brief consultation times prevent them from fully explaining every family planning option to each patient. Without a dedicated digital interface, such as a local clinic kiosk, patients have no clear way to review this critical information before their appointments. As outlined in Table I, resolving this specific bottleneck demonstrates the practical need for an AI assistant to support initial patient education.

TABLE I. PROGRESSIVE PROBLEM NARROWING

Level	**Problem Definition**
	Group Output
Broad domain	Healthcare information access gaps in the Philippines
SDG-related issue	Reproductive health and family planning clinical services
Target users	Patients, young couples, and mothers visiting Rural Health Units and Barangay Health Stations, alongside assisting healthcare workers
Narrow problem	Manual cross-referencing of DOH documents is error-prone
Proposed chatbot focus	Retrieving eligibility criteria and screening checklists

III. PRIMARY RESEARCH QUESTION

To ensure a rigorous evaluation of Retrieval-Augmented Generation [5], the proposed architecture is anchored by a single core inquiry. The primary question asks, "How effectively can a RAG-based chatbot retrieve and synthesize accurate, guideline-compliant answers for patients querying official Department of Health family planning documents?" This specific question is highly appropriate, focused, and measurable. It is appropriate because it directly tests the architecture's ability to safely bridge the clinical information

bottleneck. The scope remains tightly focused by deliberately restricting the knowledge base strictly to family planning manuals rather than open-ended maternal care. Finally, the question is highly measurable because the research relies on strict quantitative scoring to evaluate the system's safety and its ability to mitigate hallucinated medical advice. The evaluation framework compares the generated output directly against the original reference passages, establishing concrete performance metrics rather than relying on subjective user feedback.

IV. Initial Knowledge-Base Data Sources

Four authoritative sources establish the highly-curated knowledge base for the planned pipeline, detailed in Table II. Curating a strict, high-signal dataset rather than scraping general web results is a deliberate engineering choice to minimize the risk of the language model hallucinating. The Philippine Family Planning Handbook and the Clinical Standards Manual provide the core local guidelines for contraceptive counseling and surgical screening [6], [7].

These local protocols are cross-referenced against the global baseline provided by the WHO Medical Eligibility Criteria [4]. Finally, administrative policies, specifically PhilHealth circulars regarding benefit packages, contain the mandatory intake checklists required before services can be rendered [8]. Confining the knowledge base exclusively to these verified, multi-hundred-page manuals provides the architecture with a highly reliable foundation to return safe, conflict-free medical guidance.

TABLE II. Knowledge-base Sources

Source name	**Source Details**					
	Document types	***Relevance***	***Availability***	***Reliability***	***Gathered by***	***Date gathered***
Philippine Family Planning Handbook (DOH)	Clinical handbook	Local contraceptive counseling rules	Publicly downloadable PDF	Official DOH standard	Alexsandra Duhac	Sep. 20, 2026
Clinical Standards Manual on Family Planning (DOH)	Standards manual	Surgical screening and provider protocols	Publicly downloadable PDF	Official DOH standard	Anthony S. Mijares	Sep. 17, 2026
WHO Medical Eligibility Criteria (5th Ed.)	Global clinical guideline	Baseline medical contraindications	Publicly downloadable PDF	International medical standard	Jude Emmanuele D. Pineda	Sep. 17, 2026
PhilHealth Family Planning Circulary	Administrative policy	Patient intake and financial coverage	Publicly downloadable PDF	Official government policy	Anthony S. Mijares	Sep. 18, 2026

V. Initial Evaluation Data Plan

Deriving the test questions directly from the Department of Health manuals, rather than relying on generic open-source medical datasets, ensures the evaluation strictly measures compliance with local Philippine healthcare policy. To thoroughly evaluate pipeline capabilities, the evaluation taxonomy (detailed in Table III) spans multiple query types, ranging from direct factual lookups to complex eligibility cross-referencing. Manually mapping each test query to specific supporting passages within the official manuals provides an absolute ground truth for scoring.

Processing queries efficiently relies on dense vector embeddings [10] to capture the underlying semantic intent behind a patient's natural language question. This semantic mapping is crucial because it allows the system to accurately retrieve the correct clinical guidelines even if the patient lacks precise medical terminology. Generation quality then undergoes scoring on faithfulness and correctness against that retrieved context [12]. Finally, deliberately including out-of-scope questions verifies the architecture can safely refuse unsupported patient queries.

TABLE III. Table Type Styles

Query Category	**Evaluation Details**	
	Example Patient Questions	***Expected Behavior***
Direct Lookup	"How long does a subdermal implant last?"	Retrieve specific duration directly from DOH Handbook.
Eligibility Cross-Reference	"I have high blood pressure. Can I take oral contraceptive pills?"	Identify WHO MEC Category 3/4 contraindication.
Procedural or Intake	"What PhilHealth documents do I need for a free IUD insertion?"	Extract intake requirements from PhilHealth Circular.
Out-of-Scope (Safety)	"Can you diagnose why my newborn baby has a high fever?"	Safely refuse to answer; direct patient to a pediatrician.

VI. Possible Language Model and Justification

Selecting an appropriate generation model requires balancing deployment feasibility against the context window capacity required for Retrieval-Augmented Generation. Llama-3-8B-Instruct serves as the proposed generation engine [13]. Beyond its ability to run entirely on local clinic hardware, its native 8,192-token context window is strictly critical for this architecture. This capacity guarantees that multiple dense pages of retrieved Department of Health guidelines can be injected simultaneously into the prompt without truncation, ensuring the model synthesizes responses exclusively from complete reference materials rather than its internal weights [14].

Operating strictly as a local reading-comprehension engine, executing Llama-3 on-site requires no internet connection or recurring cloud API fees, unlike commercial alternatives such as GPT-4o-mini [15] or Gemini 2.0 Flash [16]. Furthermore, local execution prevents patient queries from being transmitted to external servers, strictly preserving data privacy and making the tool viable for resource-constrained clinics.

VII. Group Reflection

Balancing technical constraints with real-world patient needs required narrowing a massive healthcare issue into a specific, solvable bottleneck. Defining exactly why this

problem remains sufficiently narrow and manageable for a research team highlights the fundamental logic behind the proposed architecture.

A. Anthony Salva Mijares

Designing this architecture required pinpointing the exact moment where the current healthcare information system breaks down. Rather than attempting to overhaul how rural clinics operate, the focus remains strictly on the clinic waiting room, where unassisted patients struggle to understand complex Department of Health guidelines before their consultations. Framing this barrier as a targeted semantic retrieval challenge gave the research a clear, realistic direction.

This precise focus is exactly what makes the problem sufficiently narrow and manageable for student researchers. Because the objective avoids deploying a language model that generates independent medical diagnoses, the architecture bypasses the massive safety risks of open-ended clinical chatbots. By restricting the system to finding and synthesizing the correct paragraphs from official manuals, the research team can successfully build, evaluate, and test the architecture safely using standard computing resources.

B. Alexsandra Duhac

Focusing on the end-user experience, I realized how severely dense bureaucratic manuals and complex health guidelines alienate everyday patients. During brief consultation times at community facilities, rural healthcare workers often lack the time to comprehensively explain every available family planning option. Consequently, young couples and mothers frequently struggle to independently verify whether they meet specific clinical eligibility criteria or qualify for PhilHealth benefit coverage. Expecting unassisted patients to manually navigate hundreds of pages of technical Department of Health manuals is completely impractical. This project made me understand that reproductive health rights are heavily restricted in practice if patients cannot easily interpret the guidelines governing their care. Ensuring accessible health information is a vital requirement for translating global mandates under UN Sustainable Development Goal 3 into effective local care. Without clear administrative guidance, patients often endure unnecessary clinic visits only to discover they are ineligible for their chosen contraceptive method.

Implementing a Retrieval-Augmented Generation (RAG) architecture offers an effective computational mechanism to bridge this information gap. By mapping natural language queries directly to official government documents, the chatbot delivers verified procedural guidance on Medical Eligibility Criteria and PhilHealth intake steps. Integrating a locally hosted language model ensures that patient data privacy is strictly preserved while synthesizing retrieved passages into coherent responses. Crucially, the architecture mitigates medical hallucinations by grounding every answer in authorized clinical manuals and safely refusing diagnostic requests. Ultimately, this research demonstrates how AI can empower patients to make informed, independent reproductive choices before even arriving at a facility.

C. Jude Emmanuele Pineda

By converting complex Department of Health guidelines into grounded, verifiable answers, the architecture expands access to family planning information without assuming diagnostic responsibilities. System safety relies on strict context-retrieval mechanisms and active guardrails that reject clinical or out-of-scope prompts. Bounding the knowledge base to official reproductive health protocols circumvents the risks of general medical conversational models, creating a standardized framework to quantitatively evaluate context alignment and retrieval precision.

REFERENCES

[1] United Nations. "Goal 3: Ensure healthy lives and promote well-being for all at all ages." un.org. https://sdgs.un.org/goals/goal3

[2] Responsible Parenthood and Reproductive Health Act of 2012, Rep. Act No. 10354, Official Gazette of the Republic of the Philippines (2012). https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/2/51265

[3] Universal Health Care Act, Rep. Act No. 11223, Official Gazette of the Republic of the Philippines (2019). https://elibrary.judiciary.gov.ph/thebookshelf/showdocsfriendly/2/86448

[4] Medical Eligibility Criteria for Contraceptive Use, 5th ed., World Health Organization, Geneva, Switzerland, 2015. https://www.who.int/publications/i/item/9789241549158

[5] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in Proc. 34th Int. Conf. Neural Inf. Process. Syst. (NeurIPS), vol. 33, 2020, pp. 9459–9474. doi: 10.5555/3495724.3496517

[6] Department of Health, Philippine Family Planning Handbook, 2023 ed. Manila, Philippines: Department of Health, 2023. Available: https://www.who.int/philippines/internal-publications-detail/the-philippine-family-planning-handbook-(2023)-edition

[7] Department of Health, Clinical Standards Manual on Family Planning. Manila, Philippines: Department of Health, 2014. Available: https://platform.who.int/docs/default-source/mca-documents/policy-documents/operational-guidance/PHL-RH-32-01-OPERATIONALGUIDANCE-2014-eng-Clinical-Standards-Manual-Family-Planning.pdf

[8] Department of Health, Guidelines on the Implementation of the Family Planning Program, DOH Administrative Order No. 2026-0006, Manila, Philippines, 2026

[9] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Proc. 31st Int. Conf. Neural Inf. Process. Syst. (NIPS)*, 2017, pp. 5998–6008. doi: 10.5555/3295222.3295349.

[10] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in Proc. 2019 Conf. Empirical Methods Natural Lang. Process. (EMNLP), Hong Kong, China, 2019, pp. 3982–3992. doi: 10.18653/v1/D19-1410.

[11] J. Johnson, M. Douze, and H. Jégou, "Billion-scale similarity search with GPUs," IEEE Trans. Big Data, vol. 7, no. 3, pp. 535–547, July 2021. doi: 10.1109/TBDATA.2019.2921572.

[12] S. Es, J. James, L. Espinosa-Anke, and S. Schockaert, "RAGAS: Automated evaluation of retrieval augmented generation," in Proc. 18th Conf. Eur. Chapter Assoc. Comput. Linguistics (EACL): Syst. Demonstrations, St. Julian's, Malta, 2024, pp. 150–158. doi: 10.18653/v1/2024.eacl-demo.16.

[13] Ollama, "Ollama: Get up and running with large language models," 2024. . Available: https://ollama.com/

[14] Meta, "Introducing Meta Llama 3: The most capable openly available LLM to date," Meta AI Blog, Apr. 18, 2024. Available: https://ai.meta.com/blog/meta-llama-3/

[15] OpenAI, "GPT-4o mini: Advancing cost-efficient intelligence," OpenAI Blog, July 18, 2024. Available: https://openai.com/index/gpt-4o-mini-advancing-cost-efficient-intelligence/

[16] Google, "Introducing Gemini 2.0: Our new AI model for the agentic era," Google DeepMind Blog, Dec. 11, 2024. Available: https://blog.google/technology/google-deepmind/google-gemini-ai-update-december-2024/